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SwiftCart Week 1 Report

Quantifying the Bleed: Financial Impact of Manual Inventory Planning

SwiftCart operates a network of dark stores where inventory decisions directly affect profitability and customer satisfaction. Currently, inventory procurement relies on manual demand estimation by store managers. This analysis was conducted to quantify the financial impact of these decisions by measuring losses caused by overstocking (spoilage) and understocking (stockouts).

The results show that inaccurate demand forecasting leads to significant inventory wastage and missed sales opportunities. The analysis establishes a baseline financial bleed and demonstrates why manual inventory planning is insufficient for a fast-commerce business.

1. Objective

The objective of this analysis was to:

- Identify inventory wastage caused by overstocking.
- Identify revenue loss caused by stockouts.
- Quantify the financial impact of these inefficiencies.
- Establish a data-driven baseline for future inventory optimization.

2. Dataset Overview

Dataset contains operational information across multiple stores and product categories including:

- Units Ordered
- Units Sold
- Units Spoiled
- Stockout Events
- Peak Hour Stockouts
- Estimated Lost Sales

The categories included in the sample dataset are:

- Produce
- Dairy
- Bakery
- Pantry

3. Data Quality Assessment

Before conducting the analysis, the dataset was inspected using:

- head()
- info()
- describe()
- isna()

No missing values or structural inconsistencies were identified in the sample dataset. Therefore, no additional data cleaning was required.

4. Analysis & Findings

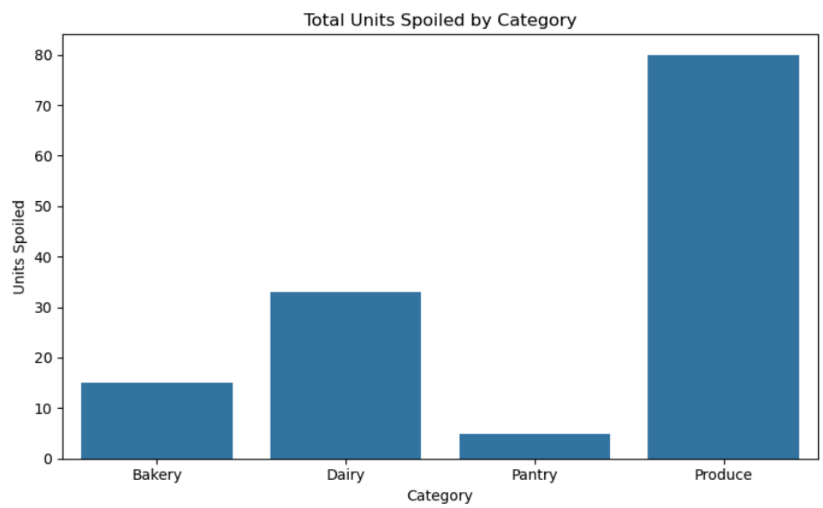
4.1 Product Wastage Analysis

Graph: Total Units Spoiled by Category

The spoilage analysis revealed the following wastage volumes:

Results:

Category	Units Spoiled
Produce	80
Dairy	33
Bakery	15
Pantry	5



Observation

Produce contributes the largest amount of spoilage, accounting for 80 spoiled units. This indicates that managers are consistently overestimating demand for highly perishable products.

Business Impact

Overstocking increases inventory carrying costs and directly reduces profitability because spoiled inventory cannot generate revenue.

4.2 Financial Cost of Spoilage

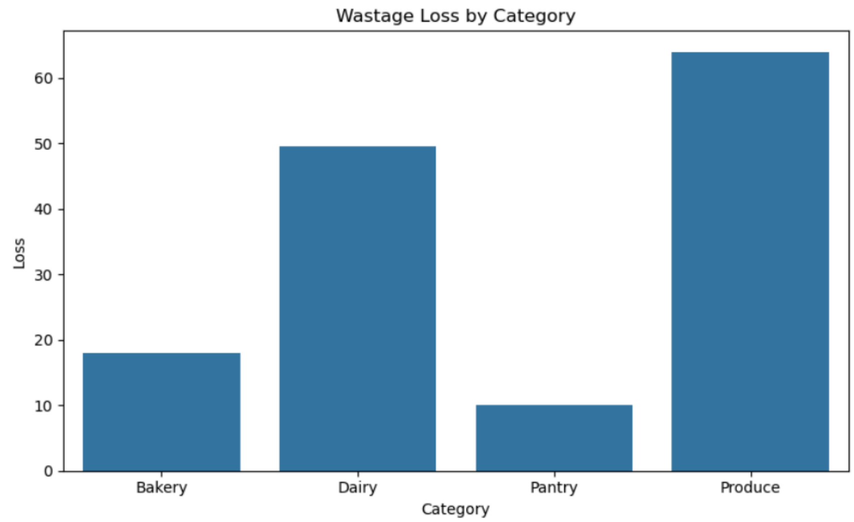
Graph: Wastage Loss by Category

Wastage cost was calculated using:

Wastage Loss = Units Spoiled × Unit Cost

Results:

Category	Wastage Loss
Produce	₹64.00
Dairy	₹49.50
Bakery	₹18.00
Pantry	₹10.00



Total Wastage Loss

₹141.50

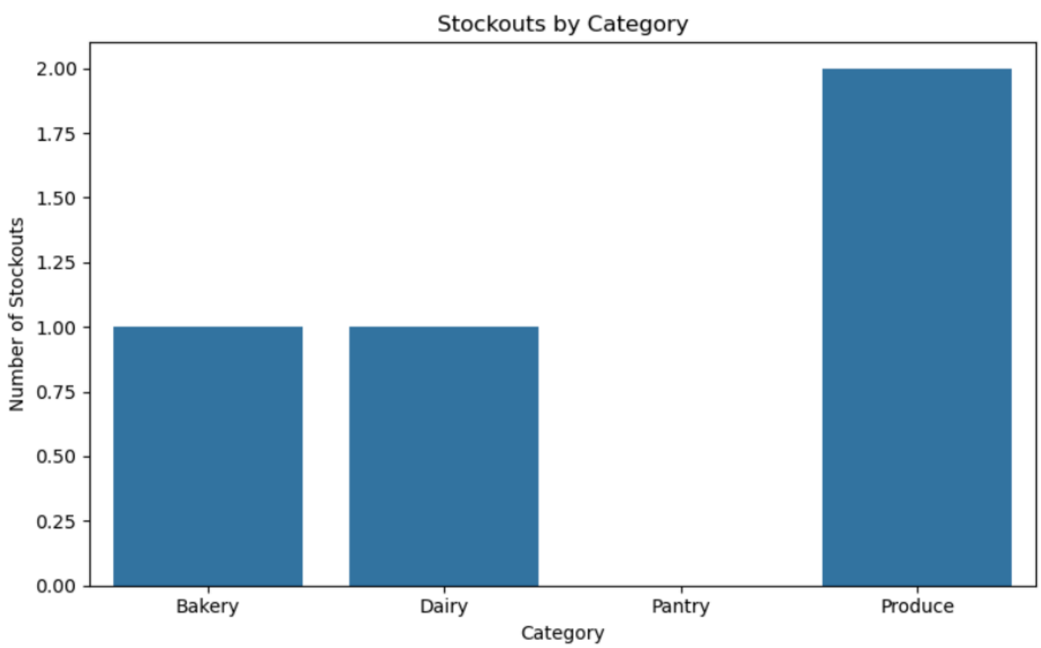
Observation

Produce generated the highest financial loss despite having a relatively low unit cost because of the large spoilage volume. Dairy generated the second-highest wastage loss.

Business Impact

The company is spending money on inventory that never reaches customers, reducing operating margins.

4.3 Stockout Analysis



The dataset recorded 4 stockout events.

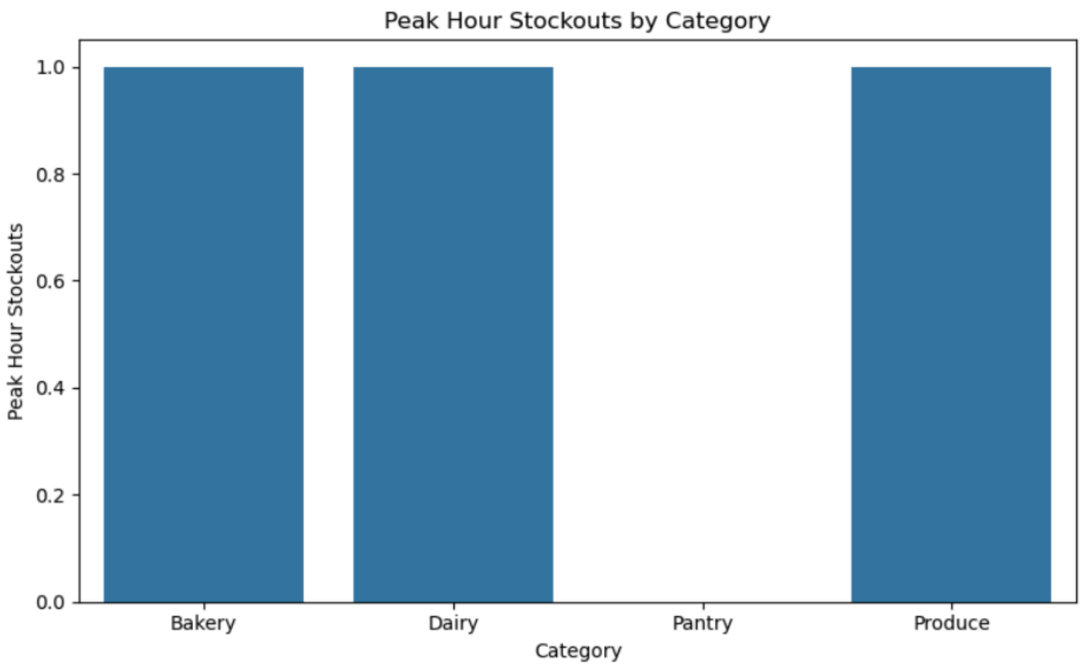
Observation

Stockouts indicate that inventory levels were insufficient to satisfy customer demand.

Business Impact

When customers cannot purchase desired products, SwiftCart loses immediate revenue and risks losing customers to competitors.

4.4 Peak-Hour Stockout Analysis



The dataset recorded 3 peak-hour stockouts.

Observation

Peak-hour stockouts are particularly damaging because they occur during periods of maximum customer demand.

Business Impact

Revenue losses during peak hours have a greater impact than losses during normal operating periods because customer purchase intent is highest.

4.5 Revenue Lost Due to Stockouts

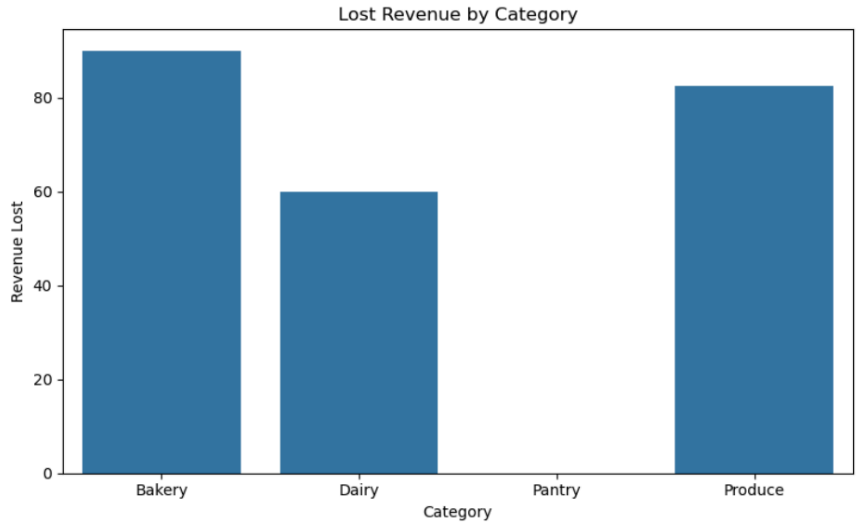
Graph: Lost Revenue by Category

Lost revenue was calculated using:

Lost Revenue = Estimated Lost Sales × Retail Price

Results:

Category	Lost Revenue
Bakery	₹90.00
Produce	₹82.50
Dairy	₹59.85
Pantry	₹0.00



Total Lost Revenue

₹232.35

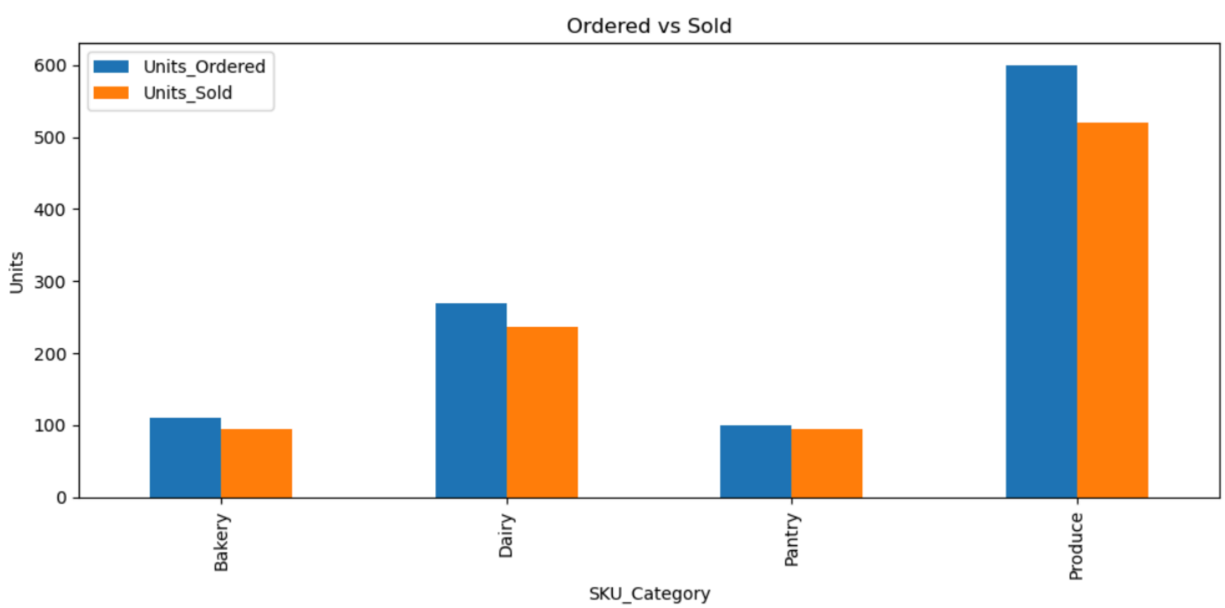
Observation

Bakery products generated the highest lost revenue despite relatively low spoilage. This indicates that demand for bakery products exceeded available inventory.

Business Impact

Understocking directly reduces revenue and negatively affects customer experience.

4.6 Ordered vs Sold Analysis



The comparison between inventory ordered and inventory sold reveals forecasting inaccuracies across all categories.

Observation

- Produce: 600 units ordered vs 520 units sold
- Dairy: 270 units ordered vs 237 units sold
- Bakery: 110 units ordered vs 95 units sold
- Pantry: 100 units ordered vs 95 units sold

Large gaps between ordered and sold quantities indicate systematic overstocking.
At the same time, stockout events occurred in several categories, proving that managers simultaneously overstock some products while understocking others.

Business Impact

This is clear evidence that manual demand estimation is unable to consistently predict customer demand patterns.

5. Financial Impact Summary

Total Wastage Loss	Total Revenue Lost Due to Stockouts	Combined Financial Loss
₹141.50	**₹232.35**	**₹373.85**

Observation

Lost revenue from stockouts is significantly higher than the direct cost of spoilage.
This suggests that understocking may be creating a larger business problem than overstocking in the sample dataset.

6. Conclusion

The analysis demonstrates that SwiftCart's current inventory planning process creates losses through both overstocking and understocking.

Key findings include:

- Produce generates the highest spoilage volume.
- Bakery products generate the highest lost revenue.
- Four stockout events occurred during the observation period.
- Three stockouts occurred during peak-demand hours.
- Total quantified loss in the sample dataset equals ₹373.85.

The Ordered vs Sold analysis confirms that manual inventory forecasting fails to accurately match customer demand.
Therefore, the evidence strongly supports the adoption of a data-driven forecasting system capable of reducing spoilage, minimizing stockouts, and improving inventory efficiency across the network.